MONTHLY TRENDS & BRAND COMPARISONS

📊 1. Total Car Sales (2022–2023)

- *What it Shows:* A KPI card summarizing the total number of cars sold over the two-year period.
- *Key Insight*: **24,000** cars were sold from 2022 to 2023.
- *Business Takeaway*: This figure reflects the overall performance of the business during the selected timeframe and sets a baseline for evaluating monthly and brand-specific sales contributions.

📈 2. Car Sales Over Time (Line Chart)

- *What it Shows*: Monthly trend of total car sales from January 2022 to October 2023.
- *Key Insights*:
    - Peaks observed in **May 2023 (585 units)** and **August 2023 (500 units)**.
    - Notable drops occurred in **April 2022 (270 units)** and **October 2022 (305 units)**.

Business Takeaway: The chart reveals that peak sales occur in May (585 units) and August 2023 (560 units), suggesting strong consumer demand in late spring and summer - possibly tied to new model releases, seasonal trends, or strategic promotions.

Major dips are seen in April 2022 (270 units), October 2022 (305 units), and September 2023 (325 units), indicating potential seasonal slowdowns or market factors worth investigating.

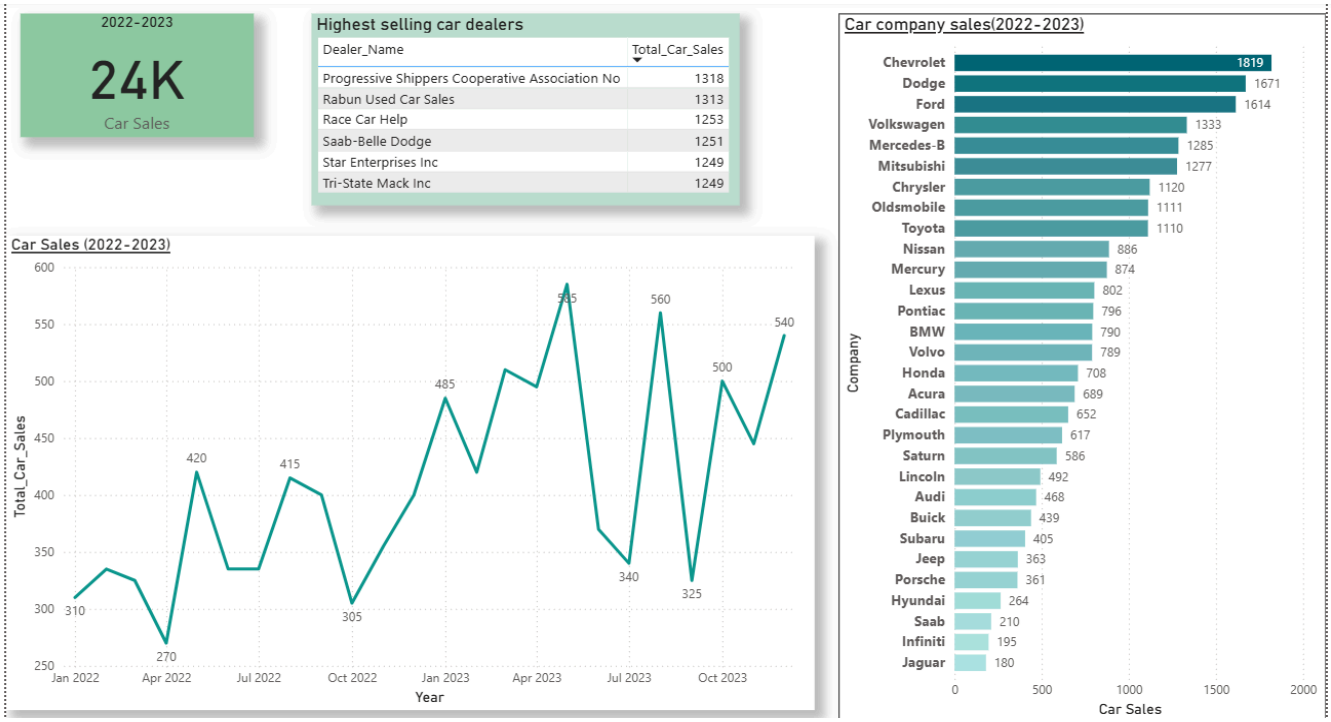
- These insights can guide forecasting, marketing efforts, and inventory allocation, helping car companies align resources with predictable demand cycles.

🏆 3. Highest Selling Dealers (Table)

- *What it Shows:* Top-performing car dealerships by total sales volume.
- *Key Insights:*
    - **Progressive Shippers Cooperative Association No** leads with **1,318 sales**, followed closely by **Rabun Used Car Sales (1,313)** and **Race Car Help (1,253)**.
- *Business Takeaway*: These dealers are outperforming the rest, indicating strong local market presence or better sales strategies. This could inform partnership or benchmarking efforts.

🚗 4. Car Company Sales Comparison (Bar Chart)

- *What it Shows*: Total car sales by company/brand across the two years.
- *Key Insights:*
    - **Chevrolet** dominates with **1,819 units**, followed by **Dodge (1,671)** and **Ford (1,614)**.
    - Brands like **Jaguar, Infiniti, and Saab** recorded the lowest sales.
- *Business Takeaway*: Chevrolet, Dodge, and Ford are market leaders. Brands with lower sales could either reflect niche targeting or underperformance, prompting further investigation.



GEOSPATIAL CAR SALES ACROSS THE US

🌎 Geospatial Car Sales Insight

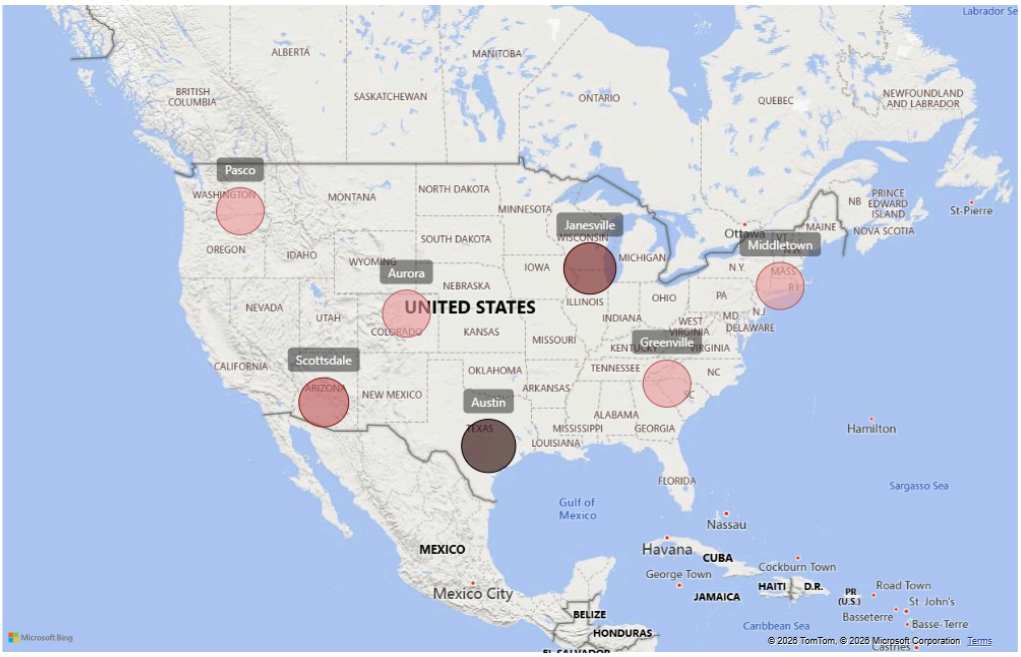
This map illustrates the **total car sales distribution across dealer regions** in the U.S. for the period under review. Key observations:

- **Austin** leads in total sales with **4,155 units**, standing out significantly from other regions. This may indicate a larger market demand, better-performing dealerships, or targeted marketing efforts in Texas.
- **Janesville (3,821)** and **Scottsdale (3,433)** follow as strong-performing regions, possibly reflecting high consumer activity or favorable economic factors.
- On the lower end, **Aurora (3,130)** and **Pasco (3,131)** recorded the least sales, which could signal lower dealership activity, smaller populations, or market saturation.
- The visual sizing and shading of the bubbles on the map offer a quick grasp of regional performance, with darker/larger circles denoting higher sales.

Business Implication:

Regions like **Austin, Janesville, and Scottsdale** may serve as benchmarks for expanding best practices or allocating more inventory. Meanwhile, lower-performing regions warrant further investigation such as; are there operational challenges, weaker brand penetration, or economic barriers?

Car Sales by Region Distribution



Dealer_Region	Total_Car_Sales
Aurora	3130
Austin	4135
Greenville	3128
Janesville	3821
Middletown	3128
Pasco	3131
Scottsdale	3433
Total	**23906**

CUSTOMER DEMOGRAPHICS

📊 Car Sales by Income Group

- **High-income customers** account for the majority of car sales with **15,370 units**, far ahead of both **low-income (5,404)** and **middle-income (3,132)** groups.
- This pattern indicates a strong purchasing power among high-income earners, likely enabling them to afford premium or multiple vehicles.

🟢 *Business Implication:* Marketing efforts and premium offerings can be more aggressively targeted at high-income regions or customer segments to maximize returns.

👫 Total Car Sales by Gender

- **Female customers dominate** the market with **19,000+ car purchases (78.63%)**, compared to **5,000+ male purchases (21.37%)**.
- This may reflect targeted marketing effectiveness, decision-making influence, or higher buyer engagement among female customers.

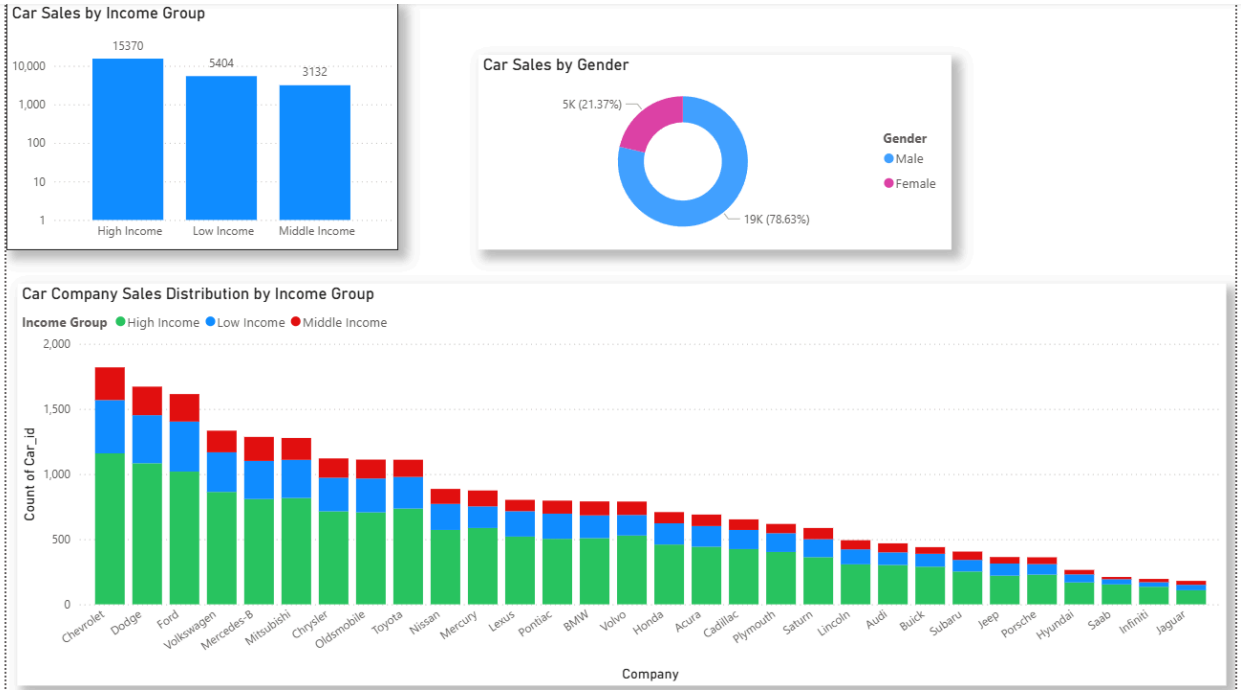
🔵 *Business Implication:* Dealerships could benefit from tailoring services, experiences, and communication more effectively toward female buyers.

🚗 Car Company Sales Distribution by Income Group

- **Chevrolet, Dodge, Ford, and Volkswagen** are the top-selling brands, especially among **high-income buyers** (green bars).
- Brands like **BMW, Mercedes-Benz, and Jaguar** also show sales primarily driven by **high-income consumers**, affirming their premium market position.
- Some brands show a more **balanced spread across income groups**, suggesting accessibility and broader appeal (e.g., Toyota, Honda).

📈 *Business Implication*:

- For luxury car brands, focus on maintaining exclusivity and high-end brand messaging.
- For mass-market brands, emphasize value and financing options that appeal across income segments.



CAR SPECIFICATIONS & PREFERENCE

⚙️ Car Sales by Transmission

- **Automatic cars** are slightly more preferred with **12.6K sales**, compared to **11.3K manual**.
- This suggests a leaning toward convenience and ease of use, possibly reflecting urban driving conditions or a younger customer base.

Business Insight: Manufacturers and dealers should maintain or increase inventory of automatic models, especially in cities or among first-time buyers.

🎨 Car Sales by Color

- **Black** remains the most popular color with **11K sales**, followed by **Pale White (8K)** and **Red (5K)**.
- This aligns with global trends where neutral and dark colors are often seen as professional and resale-friendly.

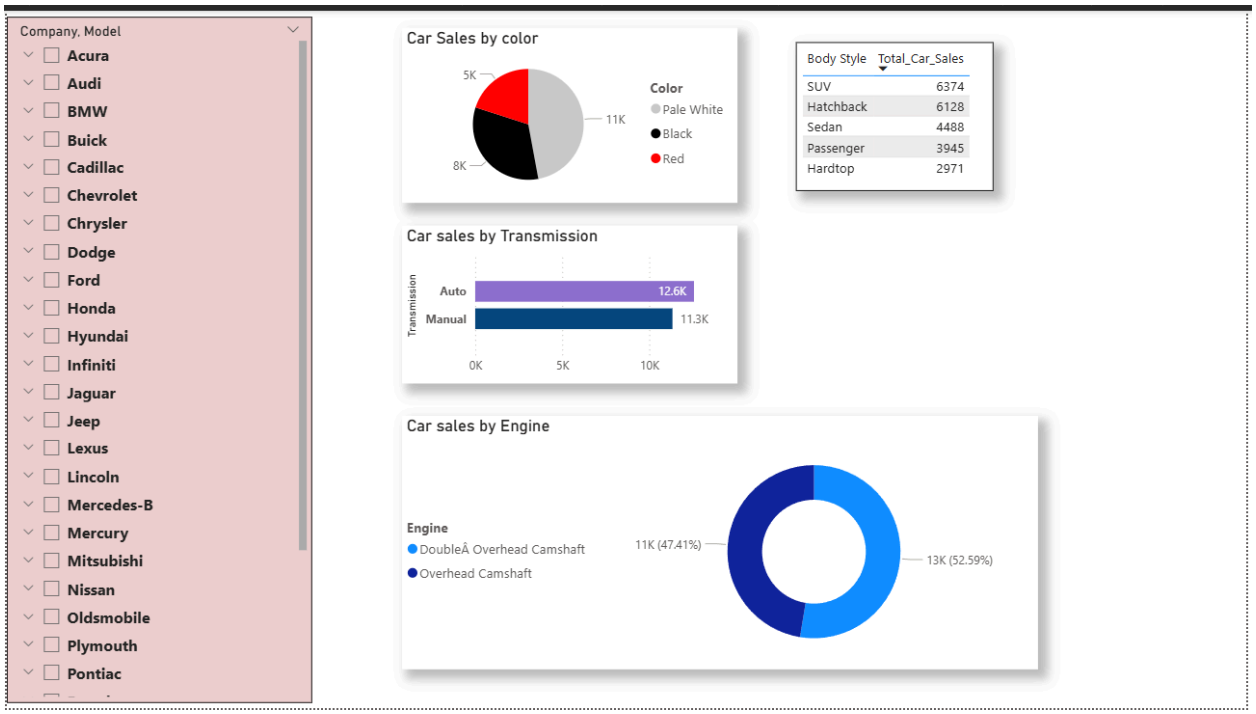
Business Insight: Car dealers can prioritize stocking black and white models while offering red and other colors as limited-edition or sport-themed options.

🛠️ Car Sales by Engine Type

- **Overhead Camshaft** engines slightly edge out with **13K sales (52.59%)**, while **Double Overhead Camshaft** follows with **11K (47.41%)**.
- This split indicates **close competition** in consumer preference, possibly influenced by brand-specific offerings or perceived performance differences.

Business Insight: Brands should continue offering both engine types but could market the **performance edge or efficiency benefits** of one over the other depending on their audience.

🚗 Car Sales by Body Style

- **SUVs (6,374)** and **Hatchbacks (6,128)** are the top-selling body types, far ahead of **Sedans (4,488)** and **Passenger/Hardtop vehicles**.
- This reflects consumer preference for space, utility, and versatility, especially in suburban or family-oriented areas.

Business Insight: Inventory planning should lean heavily into SUVs and hatchbacks. Advertising could emphasize spaciousness, safety, and fuel efficiency to target family or outdoor-oriented buyers.